

Original Article

Unlocking the gut-lung axis: Feixin decoction as a novel modulator in hypoxic pulmonary hypertension

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ABSTRACT

Background: Feixin decoction (FXD) is an effective traditional Chinese medicine prescription for treating chronic pulmonary heart disease and hypoxic pulmonary hypertension (HPH). However, the pharmacological mechanism of FXD in preventing HPH remains unclear.

Purpose: This study aimed to evaluate the preventive and therapeutic effect of FXD on HPH and confirm the association between HPH, gut microbiota, and FXD.

Methods: Multiple in vivo animal models were used, including HPH rat models, microbiota depletion models, and fecal microbiota transplantation (FMT) models. The HPH phenotype was evaluated through: right heart catheterization for hemodynamic parameters, doppler echocardiography for cardiac function assessment, hematoxylin-eosin staining for histopathological examination, and immunofluorescence labeling for specific protein expression analysis. Concurrently, transmission electron microscopy was utilized to observe the ultrastructure of the intestinal barrier, combined with immunofluorescence to examine the distribution characteristics of tight junction proteins. To elucidate the mechanism by which HPH ameliorates gut microbiota dysbiosis and associated metabolites, the study integrated 16S rRNA sequencing for microbiota composition analysis, dual-platform untargeted metabolomics for differential metabolite screening, and targeted metabolomics for quantitative validation.

Results: FXD exhibited significant therapeutic effects in HPH rats, ameliorating pulmonary vascular remodeling, attenuating right ventricular hypertrophy, reducing systemic inflammation, and restoring intestinal barrier function. Additionally, FXD partially restored intestinal ecological balance by enriching beneficial species (*Lactobacillus* and *Lactobacillus johnsonii*) while reducing pathogenic genera (*Escherichia-Shigella* and *Helicobacter rodentium*). Concurrently, FXD treatment induced favorable metabolic alterations, characterized by elevated levels of beneficial metabolites including eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), along with reduced concentrations of pro-inflammatory 5-hydroxytryptamine (5-HT). Gut microbiota depletion and fecal microbiota transplantation (FMT) studies established that FXD's therapeutic effects on HPH are mediated through gut microbiota modulation. Mechanistic investigations revealed that this protection likely involves inhibition of the TLR4/MyD88/NF-κB signaling pathway. In vitro studies further corroborated these findings, showing that FXD-enriched metabolites potently suppressed abnormal proliferation, migration and apoptosis in human pulmonary arterial smooth muscle cells (HPASMCs). Notably, EPA, the most significantly increased metabolite, specifically attenuates hypoxia-induced HPASMCs proliferation by interfering with the TLR4/MyD88/NF-κB signaling axis.

Abbreviations: FXD, Feixin decoction; HPH, hypoxic pulmonary hypertension; EPA, Eicosapentaenoic acid; DHA, Docosahexaenoic acid; COPD, chronic obstructive pulmonary disease; TCM, traditional Chinese medicine; TLR4, toll-like receptor 4; PAT, pulmonary artery acceleration time; PET, pulmonary artery ejection time; TAPSE, tricuspid annular systolic displacement; RVSP, right ventricular systolic pressure; WT%, ratio of vascular wall thickness to external diameter; PVDF, polyvinylidene fluoride.

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Conclusions: Our study confirms that FXD alleviates HPH by regulating gut microbiota and its associated metabolites and validates the potential of FXD as a gut microbiota modulator and an HPH treatment, thereby providing a new therapeutic strategy to improve treatment efficacy.

Introduction

Pulmonary hypertension (PH) is a severe vascular disease of the lungs characterized by consistently elevated pulmonary arterial pressure, leading to alterations in the pulmonary arterioles and enlargement of the right ventricle, which eventually results in right heart failure and death (Humbert et al., 2022). Current estimates indicate a prevalence of $\geq 1\%$ for PH in the global population. The prevalence is higher among individuals over the age of 65 due to the presence of both cardiac and pulmonary issues associated with PH (Nathan et al., 2019). HPH, a type of PH primarily caused by lung diseases, is the second most frequent cause of PH (Olsson et al., 2023). Gut microbiota imbalance has been identified as a major contributor to the pathogenesis and progression of PH in human patients and HPH in rodent models (Luo et al., 2023; Moutsoglou et al., 2023; Sharma et al., 2020). The gut microbiota's composition is shaped by several elements, such as diet, lifestyle, and medications (Walker et al., 2021). Compared to long-term oxygen therapy, traditional Chinese medicine (TCM) is seen as a superior treatment method for HPH (Wang et al., 2016). Growing evidence suggests that TCM exerts its therapeutic benefits through the modulation of gut microbiota and their metabolites (Gou et al., 2023). Consequently, exploring the bidirectional interactions between TCM and the gut microbiota is key for uncovering new insights into the pharmacological effects of TCM in preventing and treating HPH.

Currently, effective treatments for HPH remain limited, but TCM offers unique advantages in managing this disease. Our research team has developed a formula called FXD, which specifically targets the core pathology of "a deficiency of ancestral Qi causing fluid accumulation, phlegm, and blood stasis." Preliminary clinical studies have confirmed that FXD effectively improves the clinical symptoms of patients with cor pulmonale and reduces serum inflammatory factor levels (Gu et al., 2002; Huang et al., 2006). Nevertheless, the mechanisms underlying the therapeutic effects of FXD and its potential interactions with gut microbiota remain incompletely understood. This study employed HPH, microbiota-depleted, and FMT-treated rat models to investigate the role of FXD-microbiota crosstalk in HPH prevention.

Therefore, the findings revealed that FXD significantly improved PH phenotypes, enriched beneficial probiotics and metabolites to reverse gut microbiota dysbiosis, and enhanced systemic inflammatory factor levels and intestinal barrier function. Further research demonstrated that the therapeutic mechanism of FXD in treating HPH is partially mediated through gut microbiota-dependent downregulation of the TLR4/Myd88/NF- κ B signaling pathway. In summary, these results indicate that FXD, as a TCM, exhibits significant efficacy in prevention and treatment of HPH, with its interaction with gut microbiota playing a key therapeutic role.

Materials and methods

Analysis of chemical constituents of FXD

The FXD comprises astragalus root, white ginseng processed aconite, lepidium seed, poria, white atractylodes rhizome peach kernel, safflower, red peony root, plantago seed, ophiopogon, and anemarrhena rhizome. All ingredients comply with the 2020 edition of the "Pharmacopoeia of the People's Republic of China," purchased from the First Affiliated Hospital of Hunan University of Traditional Chinese Medicine, and authenticated by Associate Researcher Long Hongping from the same hospital. The decoction was concentrated using traditional water extraction to generate solutions with concentrations of 0.585, 1.17, and

2.34 g/ml, stored at 4 °C in a refrigerator. UHPLC/Q-TOF/MS analysis determined 46 compounds in the FXD (Fig. 1, Table S1)

Animal experiments

Seven-week-old male Sprague-Dawley (SD) rats were supplied by Hunan SJA Laboratory Animal Co., Ltd. (Animal License No.: SKXX2019-0004). The study utilized SD rats individually maintained in SPF-grade housing at Hunan University of Traditional Chinese Medicine, with ad libitum access to sterilized food and water. The experimental protocols (approval number ZYFY20211008-57) were reviewed and approved by the Institutional Animal Care and Use Committee on May 19, 2022 (ethical code ZYFY20220519-01), confirming adherence to established animal research ethics guidelines.

Hypoxic pulmonary hypertension rat model

Following the established protocol (Tian et al., 2021), we induced HPH in rats through intermittent normobaric hypoxia. The oxygen concentration in the chamber was precisely maintained at $10 \pm 0.5\%$ through dynamic nitrogen infusion. To control chamber conditions, we employed sodium lime for CO₂ absorption and anhydrous calcium chloride for humidity control. Animals underwent daily 8-hour hypoxia sessions over a 28-day period.

Gut microbiota-depleted and fecal microbiota transplantation rat model

Based on the literature (Jin et al., 2022), we used a mixed antibiotic regimen, consisting of vancomycin, neomycin sulfate, metronidazole, and ampicillin, to deplete gut microbiota in male SD rats through gavage. Rats were given either a medium dose of FXD or fecal suspension via gavage daily for 28 days, following the depletion of resident gut bacteria by antibiotics, to investigate the mechanism by which FXD may combat HPH via gut microbiota.

Echocardiographic assessment

Echocardiographic assessment was performed using the VINNO6 ultrasound Doppler imaging system (Vinnco 6, Vinnco, China) via thoracic echocardiography under light anesthesia (Yang et al., 2020). Specifically, isoflurane anesthesia (3.0 % induction, 1.5 % maintenance; Reverd Life Sciences, Shenzhen, China) was administered to rats, which were then positioned supine on a heated pad. Before imaging, depilatory cream was applied to the thoracic region to improve ultrasound access. Pulmonary artery acceleration time (PAT) and pulmonary artery ejection time (PET) were assessed using pulsed Doppler (parasternal long-axis view), and tricuspid annular systolic displacement (TAPSE) was determined via M-mode (apical four-chamber view). Measurements were taken with consistent heart rates across groups.

Hemodynamic measurements and right ventricular hypertrophy index assessment

Right ventricular systolic pressure (RVSP) was assessed in rats via right heart catheterization through the external jugular vein (Liu et al., 2021). Before the procedure, the BIOPAC MP160 multi-channel physiological recorder (USA) was calibrated, with the pressure channel selected, the catheter connected, and baseline zeroing performed. Following surgical isolation of the external jugular vein and distal ligation, a heparinized catheter was inserted through a "V"-shaped

venotomy and advanced into the right ventricle (RV) via the superior vena cava. RVSP was recorded using the physiological system. The right ventricular hypertrophy index (RVHI%) was subsequently calculated as previously described (Li et al., 2005). Post-euthanasia, cardiac tissues were excised, and the weights of the RV, left ventricle (LV), and septum (S) were measured.

Histological and immunofluorescence analysis

Based on our established protocol (Tian et al., 2021), lung tissue specimens were fixed in 4 % paraformaldehyde, sequentially dehydrated through an ethanol gradient series, and embedded in paraffin for histological processing. Morphological analysis was conducted on HE-stained sections visualized under a Leica DMI 3000B microscope. For immunofluorescence (Jiang et al., 2018), tissues underwent antigen retrieval, 0.3 % Triton X-100 permeabilization (1 h), and 5 % BSA blocking (1 h). After overnight incubation at 4 °C with primary antibodies (α -SMA: Boster, BM4172; ZO-1/Occludin: Abcam, ab221547/ab216327), sections were treated with fluorophore-conjugated secondary antibodies (2 h, dark). Fluorescence signals were documented microscopically.

RT-qPCR experiment

Total RNA was purified using the Tiangen DP419 kit, with subsequent quality control measurements obtained via micro UV spectrophotometry (Allsheng). Following the manufacturer's protocol (Novoprotein E172-YH01), cDNA synthesis and quantitative PCR analysis were performed. Gene expression quantification employed the $2^{-\Delta\Delta C_t}$ approach, utilizing primers detailed in Table S2.

Western blotting

In our earlier study (Tian et al., 2021), we outlined the western blot experimental protocols in detail. The primary antibodies applied in this research were NF- κ B p65 (Proteintech, 10745), TLR4 (Proteintech, 66350), MyD88 (Cell Signaling Technology, D80F5), and β -actin (Proteintech, 66009).

16S rRNA sequencing analysis

Fecal specimens were collected under cryogenic conditions and preserved at -80°C prior to analysis. OE Biotech (Shanghai) performed 16S rRNA sequencing following DNA extraction from samples. The V3-V4 hypervariable regions were PCR-amplified using 338F-806R primers and sequenced on an Illumina MiSeq system. After quality filtering, OTU clustering was conducted, followed by taxonomic classification against the SILVA 138 database via QIIME2's q2-feature-classifier. Subsequent bioinformatics processing was completed using the OECloud platform.

Dual platforms non-targeted sequencing analysis

Non-targeted metabolomic profiling was conducted by OE Biotechnology (Shanghai) using GC-MS and LC-MS platforms. Raw data were processed through Progenesis QI V3.0 (Nonlinear Dynamics, UK). GC-MS compounds were matched against the NIST database, while LC-MS metabolites were identified using LipidMaps (v2.3), HMDB, EMDB 2.0 and METLIN. Subsequent bioinformatics analysis was performed via the OECloud platform.

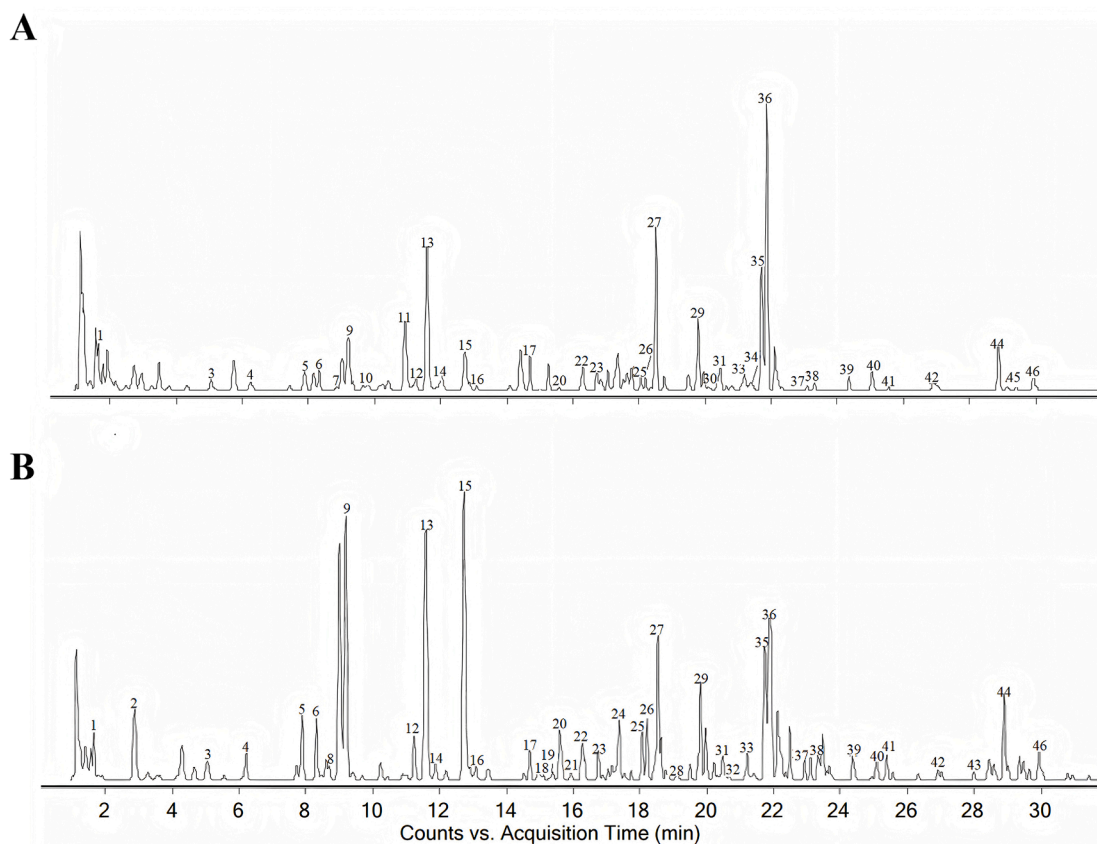


Fig. 1. Identification results of the main chemical components of FXD by UPLC-Q-TOF/MS. A: Total ion flow diagram of FXD in negative ion mode. B: Total ion flow diagram of FXD in positive ion mode.

Culture of human pulmonary artery smooth muscle cells (HPASMCs)

HPASMCs were obtained from iCell (HUM-iCell-a009) and maintained in DMEM-F12 containing 10 % fetal bovine serum (Procell, 164210), 1 % penicillin/streptomycin solution, and the purification of PASMCs were identified by immunofluorescence staining with α -SMA. We used 4–8 generations of HPASMCs in our experiment.

HPASMCs proliferation assays

HPASMCs were plated in 96-well plates (2×10^3 cells/well) and serum-starved for 24 h prior to drug treatment under normoxic or hypoxic conditions. Cell viability was determined after 24 h using the Enhanced CCK-8 kit (Elabscience, E-CK-A362). For proliferation analysis, 2×10^4 cells were grown on coverslips in 24-well plates, treated with compounds, and exposed to either normoxic or hypoxic conditions before 5-ethynyl-2'-deoxyuridine (EdU) incorporation assessment (Beyotime, C0071S).

HPASMCs migration assays

Cell migration capacity was evaluated using transwell chambers (Feng et al., 2021). Serum-starved HPASMCs (1×10^4 cells) were loaded into upper compartments, while drug-treated complete medium served as chemoattractant in lower compartments. After 24 h, membranes were cleared of non-migrated cells, with transmigrated cells undergoing methanol fixation and crystal violet staining (5 %). Migration was quantified by cell counting across multiple microscopic fields (minimum 5 fields/filter).

Cell migratory capacity was assessed through scratch wound assays (Feng et al., 2021). After achieving monolayer confluence in 6-well plates and subsequent serum deprivation (24 h), uniform wounds were introduced mechanically. Post-treatment under either hypoxic or normoxic conditions (24 h), wound closure progression was visualized and quantified by determining the relative displacement of migrating cell boundaries compared to original wound margins.

HPASMCs apoptosis assays

We seeded HPASMCs in 6-well plates (5×10^4 cells/well) and cultured them for 24 h. After drug administration, cells were maintained under normoxic or hypoxic conditions for another 24 h. Following the protocol provided with the Annexin V-FITC/PI Apoptosis Kit (Elabscience), we assessed apoptosis rates by flow cytometry (Beckman CytoFLEX S).

Statistical analysis

The quantitative data are represented as mean $\pm$ standard deviation (SD), with statistical analyses conducted using GraphPad Prism software (version 9.5). A one-way ANOVA was conducted for comparisons across multiple groups, followed by Dunnett's post hoc test. The Student's *t*-test evaluated group differences, with *p*-values below 0.05 indicating statistical significance.

Results

FXD relieves the symptoms of PH induced by hypoxia

To examine the preventive impact of FXD on HPH rats, SD rats received daily gavage doses of FXD at 5.58 g/kg (low), 11.7 g/kg (medium), and 23.4 g/kg (high), while sildenafil at 0.025 g/kg was used as a positive control. Fig. 2A shows the experimental procedure. Doppler echocardiography was employed to visualize the waveform profile of the pulmonary artery outflow (Fig. 2B). SD rats demonstrated pronounced changes in PA waveform after 4 weeks of hypoxia, confirming

the successful establishment of the hypoxic pulmonary artery pressure model. The PAT/PET ratio (a measure of RV afterload linearly related to RVSP) and TAPSE were utilized to quantify the process, and results indicated that FXD treatment reversed these changes (Fig. 2E–F). Further hemodynamic assessments showed a significant increase in RVSP in SD rats after 4 weeks of hypoxia, while FXD dose-dependently reduced this hypoxia-induced RVSP rise. (Fig. 2C,G). Right ventricular hypertrophy index results indicated that FXD improved hypoxia-induced right ventricular hypertrophy effectively (Fig. 2H). HE staining and α -SMA immunofluorescence staining revealed that pulmonary vascular wall thickness significantly increased after 28 days of chronic hypoxia, but FXD treatment greatly reduced this increase, with the medium dose demonstrating the best effect (Fig. 2D, I–J). Importantly, no adverse reactions were found in FXD-treated rats. FXD treatment had no significant impact on rat body weight (Fig. S1) or on markers of liver function and kidney function (Table S3), indicating the safety of FXD. Overall, these results indicate that FXD treatment effectively alleviates chronic hypoxia-induced PH symptoms in rats.

FXD alleviates gut microbiota dysbiosis in HPH rats

We used 16S rRNA sequencing to analyze fecal samples from rats treated with FXD for 28 days, aiming to determine if FXD affects the gut microbiota during HPH development. A Venn diagram illustrated OTU distribution across different groups (Fig. 3A). The HPH rats treated with FXD showed a notably greater number of unique OTUs than the model group, suggesting increased microbial diversity after FXD treatment.

The alpha diversity indices, Chao1 and Shannon, showed that the model group had lower community richness and diversity, whereas FXD treatment helped restore both richness and diversity (Fig. 3B–C). Additionally, the beta-diversity analysis revealed a clear separation of microbial communities within three study groups (Fig. 3D). These findings support the regulatory effect of FXD on gut microbiota diversity in rats. Further gut microbiota composition analysis revealed that FXD significantly reversed the increase in *Firmicutes*, *Spirochaetes*, *Desulfobacterota*, and *Campylobacterota* induced by hypoxia, as well as the decrease in *Bacteroidota* and *Actinobacterota*, at the phylum level (Fig. 3E). The model group exhibited a significant reduction in beneficial genera, including *Lactobacillus*, *Alloprevotella*, *Lachnospiraceae_NK4A136*, and *Romboutsia* ($p < 0.001$), while harmful genera such as *Escherichia-Shigella* and *Helicobacter* were significantly elevated ($p < 0.001$) at the genus level. Following FXD treatment, the abundance of these genera was partially or fully restored (Fig. 3F). A more granular analysis at the species level demonstrated that FXD treatment significantly enriched beneficial *Lactobacillus* species, particularly *Lactobacillus_murinus*, *Lactobacillus_reuteri*, and *Lactobacillus_johnsonii*, while concurrently decreasing the presence of harmful bacteria like *Escherichia-Shigella* and *Helicobacter_rodentium* (Fig. 3G). We used the LEfSe method to pinpoint bacterial taxa with phylogenetic enrichment that played a significant role in the β -diversity differences among the experimental groups. We conducted linear discriminant analysis (LDA) along with effect size measurements (LEfSe) to pinpoint the main bacterial taxa responsible for differences in gut microbiota composition between groups. The analysis demonstrated a distinct bacterial signature in the FXD group, predominantly represented by beneficial genera including *Lactobacillus*, *Lachnospiraceae_NK4A136*, *Alloprevotella*, and *Romboutsia*. In contrast, the model group showed an enrichment of potentially pathogenic taxa, particularly *Escherichia-Shigella* and *Helicobacter* (Fig. 3H). Subsequent in-depth analysis revealed that FXD treatment exerted a dual modulatory effect on the gut microbiota composition, demonstrating a significant upregulation of commensal probiotics, particularly *Lactobacillus_murinus* and *Lactobacillus_johnsonii* (Fig. 3I–J), concurrent with a marked suppression of pathogenic taxa, specifically *Escherichia-Shigella* and *Helicobacter_rodentium* (Fig. 3K–L). Collectively, these findings demonstrate that FXD ameliorates HPH through gut microbiota remodeling, characterized by the restoration of microbial

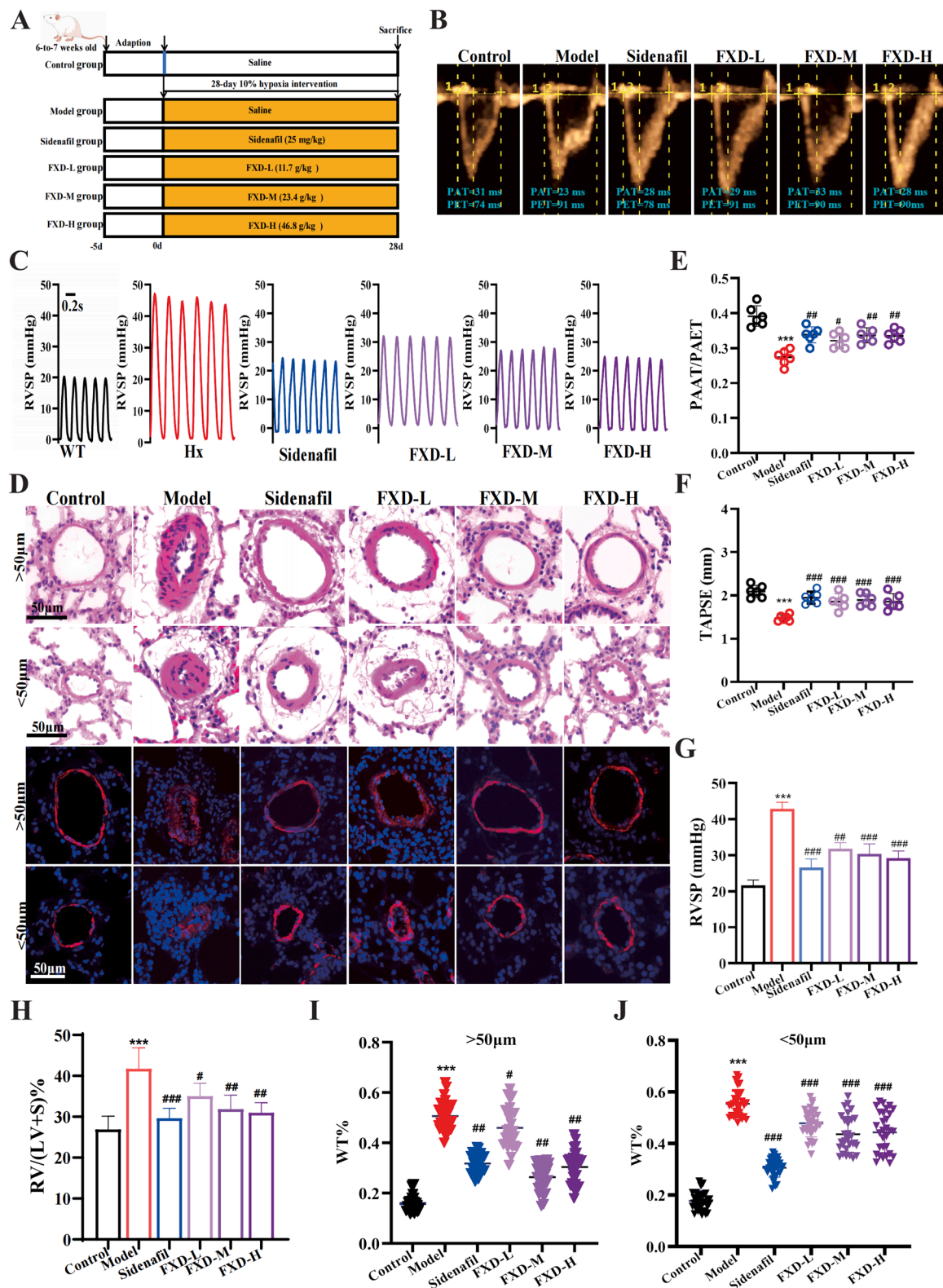


Fig. 2. FXD relieves the phenotype of pulmonary hypertension in HPH rats. **A:** Schematic diagram of the animal experiment for FXD preventive treatment in HPH rats; **B:** Doppler echocardiography measuring pulmonary artery (PA) function, showing the ratio of PAT to PET; **C:** RVSP waveform diagram; **D:** Representative H&E staining and α -SMA (red) immunofluorescence staining of PA in HPH rats treated with FXD; **E-G:** Statistical analysis of PAT/PET, TAPSE and RVSP; **H:** Statistical analysis of right ventricular hypertrophy index; **I-J:** Quantitative statistical analysis of PA vascular wall thickness. Compared to the normal group, $**p < 0.01$, $***p < 0.001$; compared to the model group, $\#p < 0.05$, $\#\#p < 0.01$, $\#\#\#p < 0.001$.

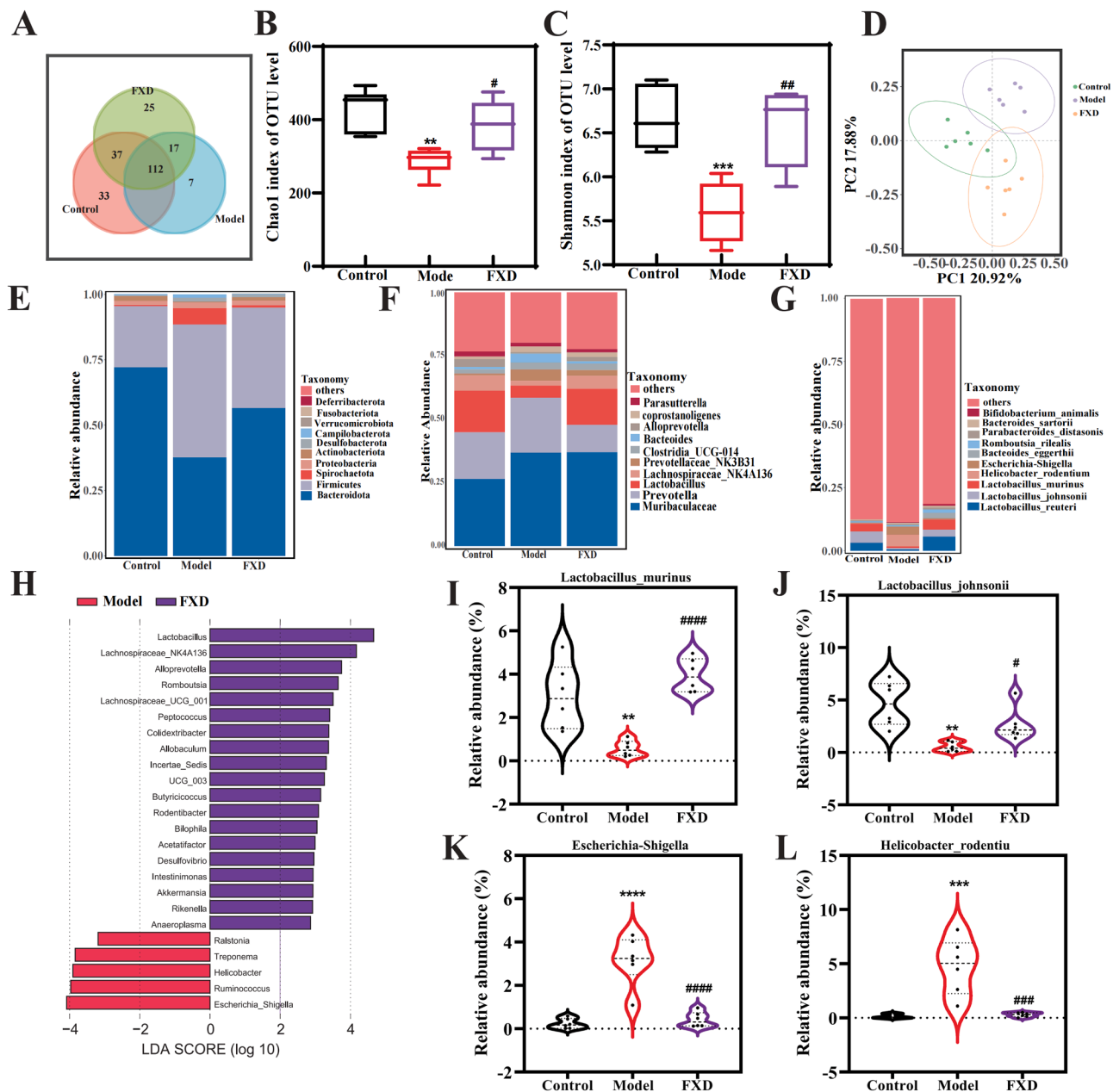


Fig. 3. FDX alleviates dysbiosis of the gut microbiota caused by HPH A: Venn diagrams of the three groups; B-C: Alpha diversity analysis of the gut microbiota using Chao1 and Shannon indices; D: Beta diversity analysis of the gut microbiota using PCoA; E-G: Analysis of microbial composition at the phylum, genus, and species levels for each group; H: Linear discriminant analysis (LDA) effect size (LefSe) analysis to identify differentially abundant bacterial taxa between groups (LDA score < 3.5); I-J: Relative abundance of significantly altered probiotics: *Lactobacillus murinus* and *Lactobacillus johnsonii*; K-L: Relative abundance of significantly altered pathogenic bacteria: *Escherichia-Shigella*, *Helicobacter rodentium*. Compared with the control group, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$; compared with the model group, # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$, #### $p < 0.0001$.

homeostasis via selective enrichment of beneficial species and specific reduction of pathogenic bacteria.

FDX improved plasma metabolites disorder in HPH rats

To investigate the role of gut microbiota-derived metabolites in health and disease regulation, we conducted comprehensive non-targeted metabolomic profiling of plasma samples across all experimental groups using both LC-MS and GC-MS platforms. Distinct inter-group separations were demonstrated by the Partial Least Squares Discriminant Analysis (PLS-DA) models built from LC-MS and GC-MS datasets, indicating significant metabolic variations among the groups. (Fig. 4A-B). Metabolomic analysis revealed significant alterations in

plasma metabolite profiles between the FDX and model groups. LC-MS analysis revealed 294 metabolites were increased and 167 were decreased in the FDX group compared to the model group, whereas GC-MS analysis identified 28 increased and 13 decreased metabolites (Fig. 4C-E). Further investigation of HPH-related differential metabolites demonstrated that the FDX treatment effectively reversed the metabolic imbalance observed in the model group, characterized by elevated harmful metabolites and reduced beneficial metabolites (Fig. 4F). According to the KEGG (Kyoto Encyclopedia of Genes and Genomes) pathway analysis, FDX notably altered the unsaturated fatty acid metabolic pathway when compared to the model group (Fig. 4G). We further performed quantitative analysis of beneficial and detrimental metabolites. As shown in Fig. 5 key beneficial

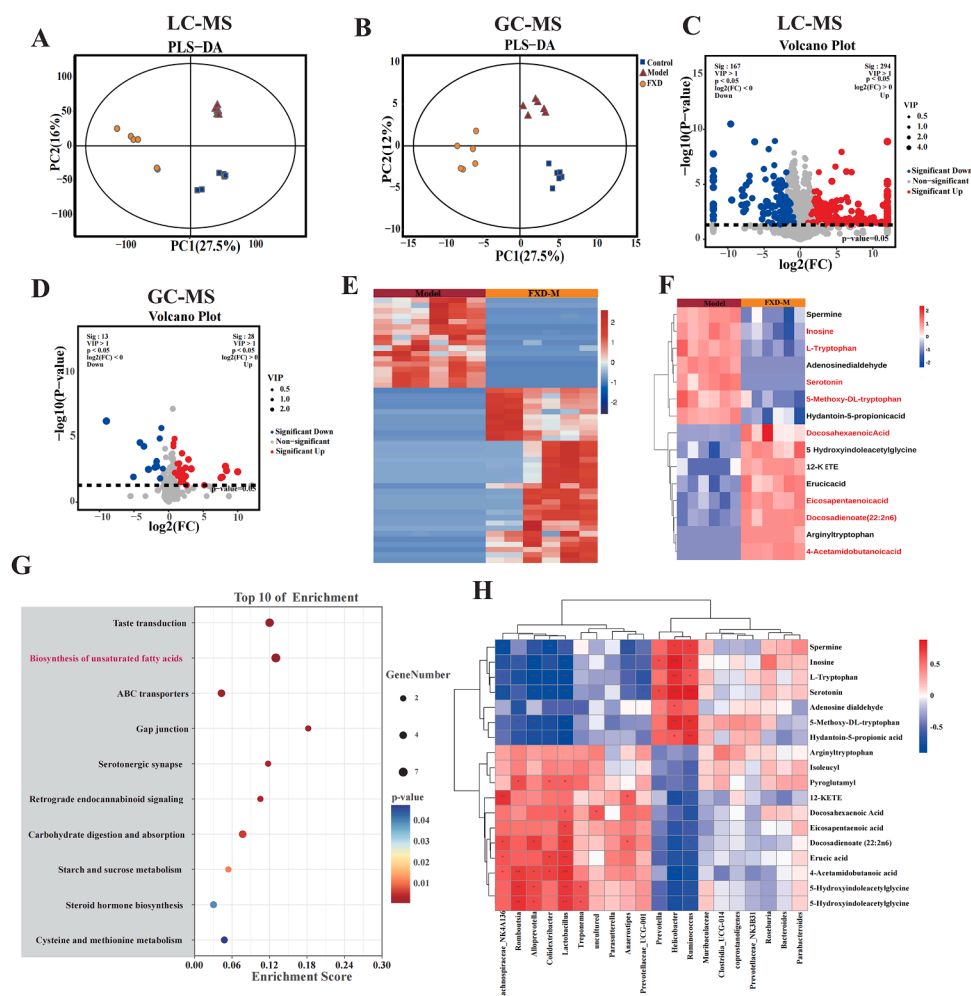


Fig. 4. FXD regulates metabolic imbalance in hypoxic pulmonary hypertension rats. (A-B) Partial Least Squares Discriminant Analysis (PLS-DA) scores among control group, Model group and FXD group. (C-D) Volcanic diagram of differentially expressed metabolites between Model group and FXD group. (E) Hierarchical cluster analysis thermogram of differential metabolite expression level between Model group and FXD group. (F) Z-score analysis thermogram of differential metabolites. (G) Metabolic pathway enrichment analysis of differential metabolites based on KEGG database. (H) Spearman correlation analysis between intestinal flora and differentially expressed plasma metabolite parameters. Correlation analysis values are expressed in particle color. Red stands for positive correlation and blue stands for negative correlation.

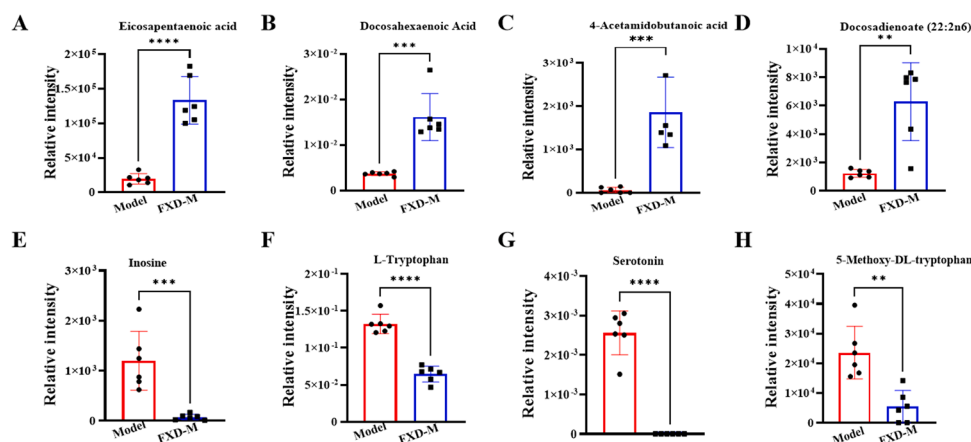


Fig. 5. Effects of FXD on representative metabolites of blood in hypoxia-induced pulmonary hypertension in rats. The relative intensities of (A) Eicosapentaenoic acid (EPA), (B) Docosahexaenoic acid (DHA), (C) 4-Acetamidobutanoic acid (GABA), (D) Docosadienoate (22:2n6), (E) Inosine, (F) L-Tryptophan, (G) Serotonin (5-HT), (H) 5-Methoxy-DL-tryptophan in model and FXD-M groups. Data were expressed as mean $\pm$ SD ($n = 6$), compared with the model group, **** $p < 0.0001$, *** $p < 0.001$, ** $p < 0.01$.

metabolites—including EPA, DHA, docosadienoate (22:2n6), and 4-acetamidobutanoic acid (GABA)—were significantly enriched after FXD treatment compared to the model group. Conversely, harmful metabolites such as serotonin, L-tryptophan, 5-methoxy-DL-tryptophan, and inosine were markedly reduced following FXD administration. Notably, emerging evidence suggests a crucial interplay among gut microbiota, unsaturated fatty acids in regulating intestinal inflammation. Recent studies have highlighted the therapeutic potential of unsaturated fatty acids in targeting the inflammatory components of PH (Massaro et al., 2024). These findings support the hypothesis that the gut microbiota-unsaturated fatty acids axis may represent a novel therapeutic target for the prevention and treatment of HPH.

We comprehensively analyzed the changed bacteria and metabolites to link these metabolites with intestinal microorganisms. Our observations indicated that the probiotics *Lachnospiraceae_NK4A136*, *Lactobacillus*, and *Alloprevotella* had a negative correlation with Serotonin and Spermine, while showing a positive correlation with 12-KETE (Fig. 4H). However, the pathogenic bacteria *Helicobacter* and *Escherichia-Shigella* were positively correlated with Inosine, L-Tryptophan, and 5-Methoxy-DL-tryptophan, but negatively correlated with 12-KETE, 4-Acetamidobutanoic acid, and 5-Hydroxyindoleacetyl-glycine. Collectively, our findings suggest that intestinal microbiota and their metabolic products may act synergistically to mediate the protective and therapeutic effects of FXD in hypoxic pulmonary hypertensive rats.

FXD improves intestinal barrier function and systemic proinflammatory factors in HPH rats

Patients with PH and rats with HPH both demonstrated abnormal intestinal barrier function. We assessed intestinal barrier permeability using fluorescein isothiocyanate (FITC)-dextran to determine whether FXD improved the intestinal barrier function in HPH rats. Compared to the model group, the FXD-M treatment group showed significantly lower levels of FITC-dextran ($p < 0.05$) (Fig. 6A). In the FXD-M treatment group, plasma bacterial lipopolysaccharide (LPS) levels, which are crucial for assessing intestinal barrier integrity, were notably lower than those in the model group ($p < 0.05$) (Fig. 6B). The molecular examination of intestinal barrier elements showed that FXD-M treatment notably increased the mRNA levels of essential tight junction proteins, ZO-1 and Occludin, in colon tissues compared to the model group ($p < 0.05$) (Fig. 6C-D). Immunofluorescence staining provided further evidence for this finding by demonstrating a significant increase in ZO-1 and Occludin protein levels in the FXD-M group ($p < 0.05$) (Fig. 6E). Ultrastructural analysis using transmission electron microscopy (TEM) provided additional evidence, showing that FXD-M treatment effectively restored the architectural integrity of the gut barrier (Fig. 6F). Given the crucial role of systemic inflammation in HPH pathogenesis, we quantified plasma levels of key proinflammatory cytokines. Rats receiving medium-dose FXD demonstrated marked reductions in proinflammatory markers, with plasma TNF- α , IL-6, and IL-1 β levels being significantly attenuated relative to untreated HPH controls ($p < 0.01$) (Fig. 6G-I).

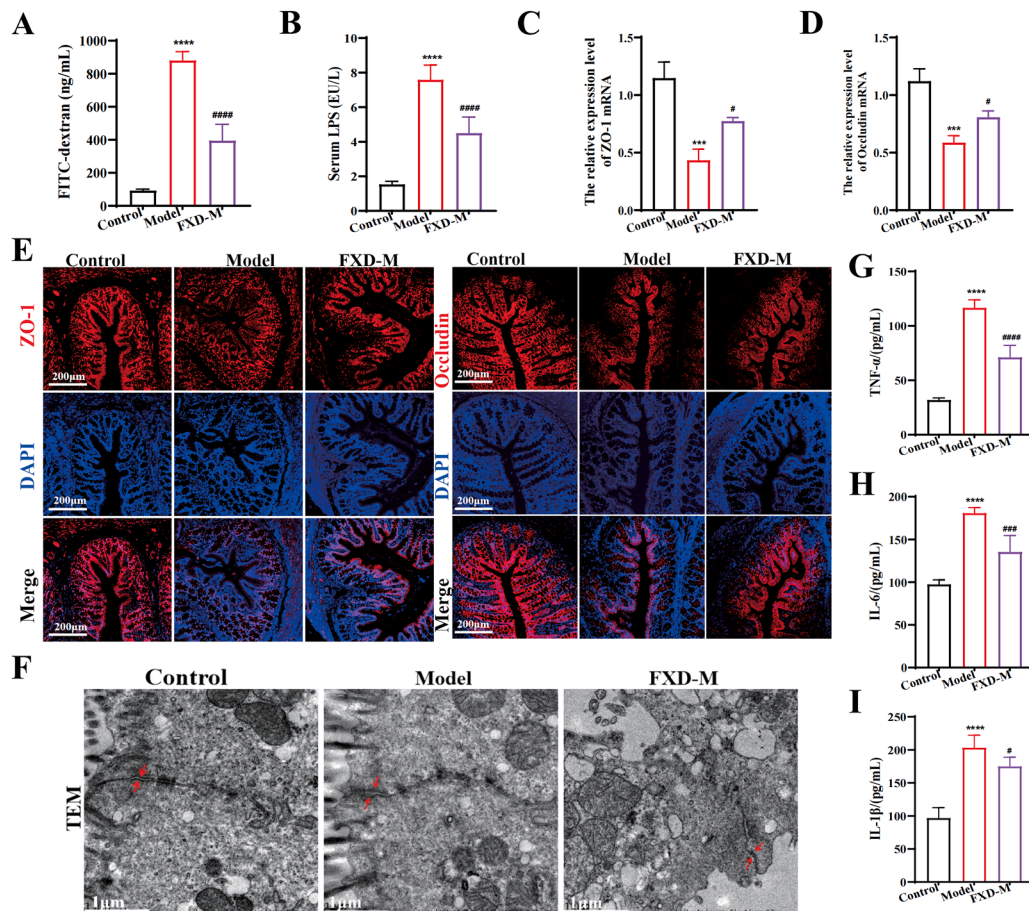


Fig. 6. FXD improves intestinal barrier dysfunction in HPH rats. A-B: Measurement of serum FITC-dextran and lipopolysaccharide (LPS) concentrations in rats; C-D: RT-qPCR analysis of the expression of intestinal barrier-related proteins ZO-1 and Occludin in rat colon tissues; E: Immunofluorescence detection of ZO-1 and Occludin in rat colon tissues; F: Representative images of intercellular junctions in rat colon tissues observed by TEM; G-I: Statistical analysis of inflammatory cytokine levels (TNF- α , IL-6, and IL-1 β) in plasma from the three groups of rats. Compared with the control group, *** $P < 0.001$, **** $P < 0.0001$; compared with the model group, # $P < 0.05$, ## $P < 0.01$, ### $P < 0.001$, #### $P < 0.0001$.

Together, these results indicate that the FXD-gut microbiome interaction protects against HPH via various pathways, such as repairing intestinal barrier integrity and reducing systemic inflammatory responses.

Gut microbiota mediates the therapeutic effect of FXD on HPH rats

We hypothesized that the gut microbiota may play a role in mediating the efficacy of FXD in treating HPH, given its contribution to the pathophysiology of HPH. To test this hypothesis, we depleted the resident microbiota in rats with HPH using antibiotics prior to FXD intervention (Fig. 7A). In the ABX+FXD-M group, both the PAT/PET ratio and the TAPSE right heart function index were notably lower than in the FXD-M group ($p < 0.05$) (Fig. 7B-D). The hemodynamic evaluations indicated that the RVSP value was notably greater in the ABX+FXD-M group compared to the FXD-M group ($p < 0.05$) (Fig. 7E). Analysis of right heart hypertrophy indicated that the ABX+FXD-M group exhibited more pronounced hypertrophy than the FXD-M group ($p < 0.01$) (Fig. 7F). The ABX+FXD-M group showed more significant pulmonary vascular remodeling than the FXD-M group, as indicated by histological analyses including HE staining and α -SMA immunofluorescence ($p < 0.01$) (Fig. 7G-I). Together, these findings propose that the therapeutic effects of FXD in HPH rats are lessened by the depletion of gut

microbiota.

We conducted FMT experiments to explore the causal role of gut microbiota in the therapeutic effects of FXD on HPH. Microbiota-depleted rats received fecal samples from either model group or FXD-treated groups, followed by hypoxia exposure (Fig. 8A). Comprehensive cardiopulmonary assessments demonstrated significant improvements in the FMT-FXD group compared to the FMT-model group. Echocardiographic analysis revealed markedly increased PAAT/PAET ratio and TAPSE values in the FMT-FXD group ($p < 0.01$) (Fig. 8B-D). Hemodynamic measurements showed substantially reduce RVSP in FMT-FXD recipients ($p < 0.01$) (Fig. 8E). Furthermore, the FMT-FXD group exhibited significant attenuation of hypoxia-induced right ventricular hypertrophy, as evidenced by the hypertrophy index (Fig. 8F). Histopathological analysis using HE staining and immunofluorescence verified that FMT-FXD treatment led to a significant reduction in pulmonary vascular wall thickness compared to the FMT-model group ($p < 0.01$) (Fig. 8G-I). These findings provide compelling evidence that the therapeutic efficacy of FXD against HPH is mediated, at least in part, through modulation of gut microbiota and associated metabolites.

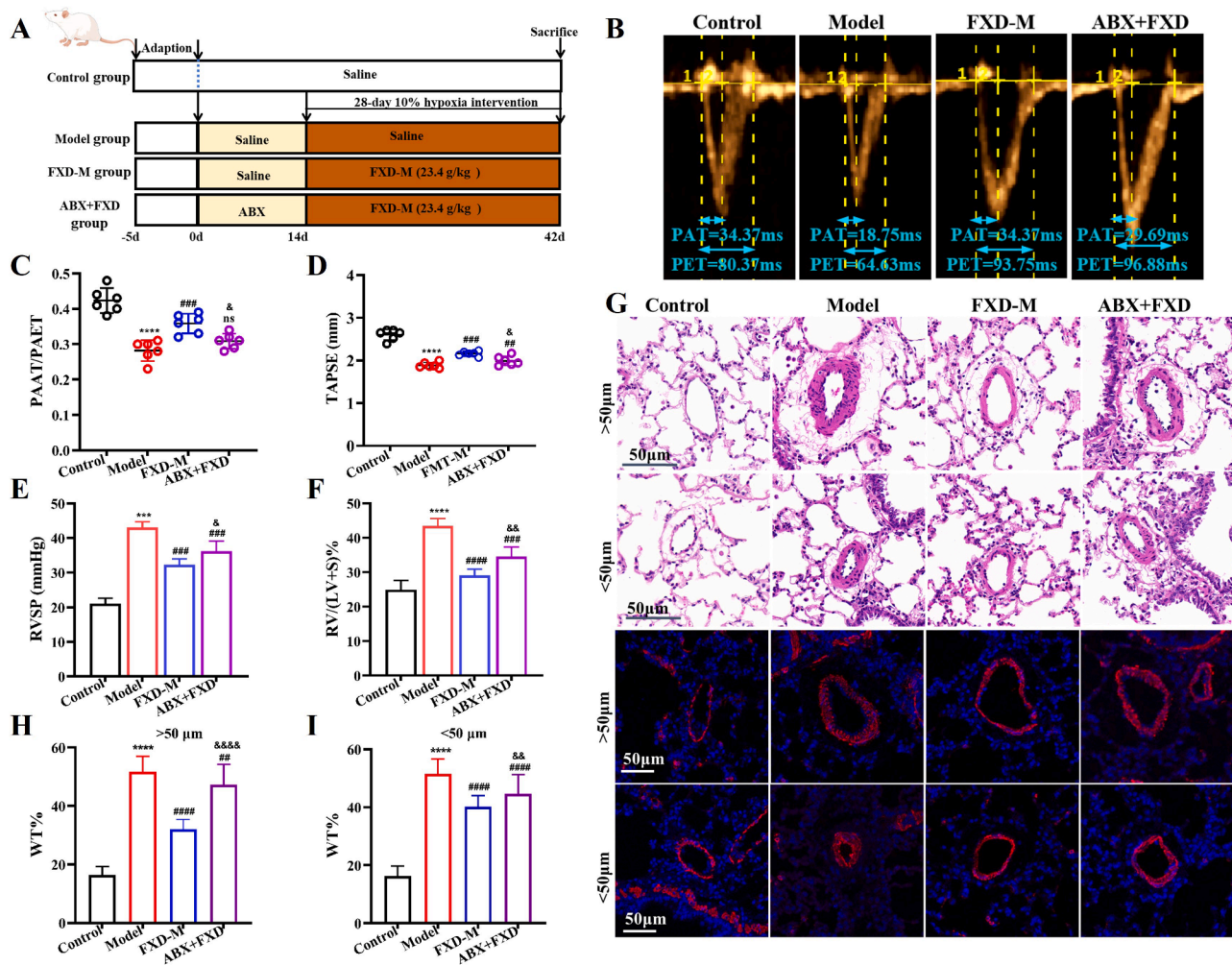


Fig. 7. The effect of gut microbiota depletion on the pulmonary vascular remodeling phenotype in HPH rats treated with FXD. A: Animal experiment design (6 rats per group), with ABX administered to rats via gavage for 14 days. B: Echocardiographic measurement of PA function showing PAT/PET ratio; C-D: Statistical analysis of PAT/PET and TAPSE; E: Statistical analysis of RVSP; F: Statistical analysis of right heart hypertrophy index; G: H&E staining and α -SMA (red) immunofluorescence staining representative images of PA in HPH rats treated with FXD, with intestinal microbiota depletion; H-I: Quantitative statistical analysis of PA vessel wall thickness. Compared to the normal group, ** $P < 0.01$, *** $P < 0.001$; compared to the model group, # $P < 0.05$, ## $P < 0.01$; compared to the FXD-M group, && $P < 0.01$, &&& $P < 0.001$; ns $p > 0.05$.

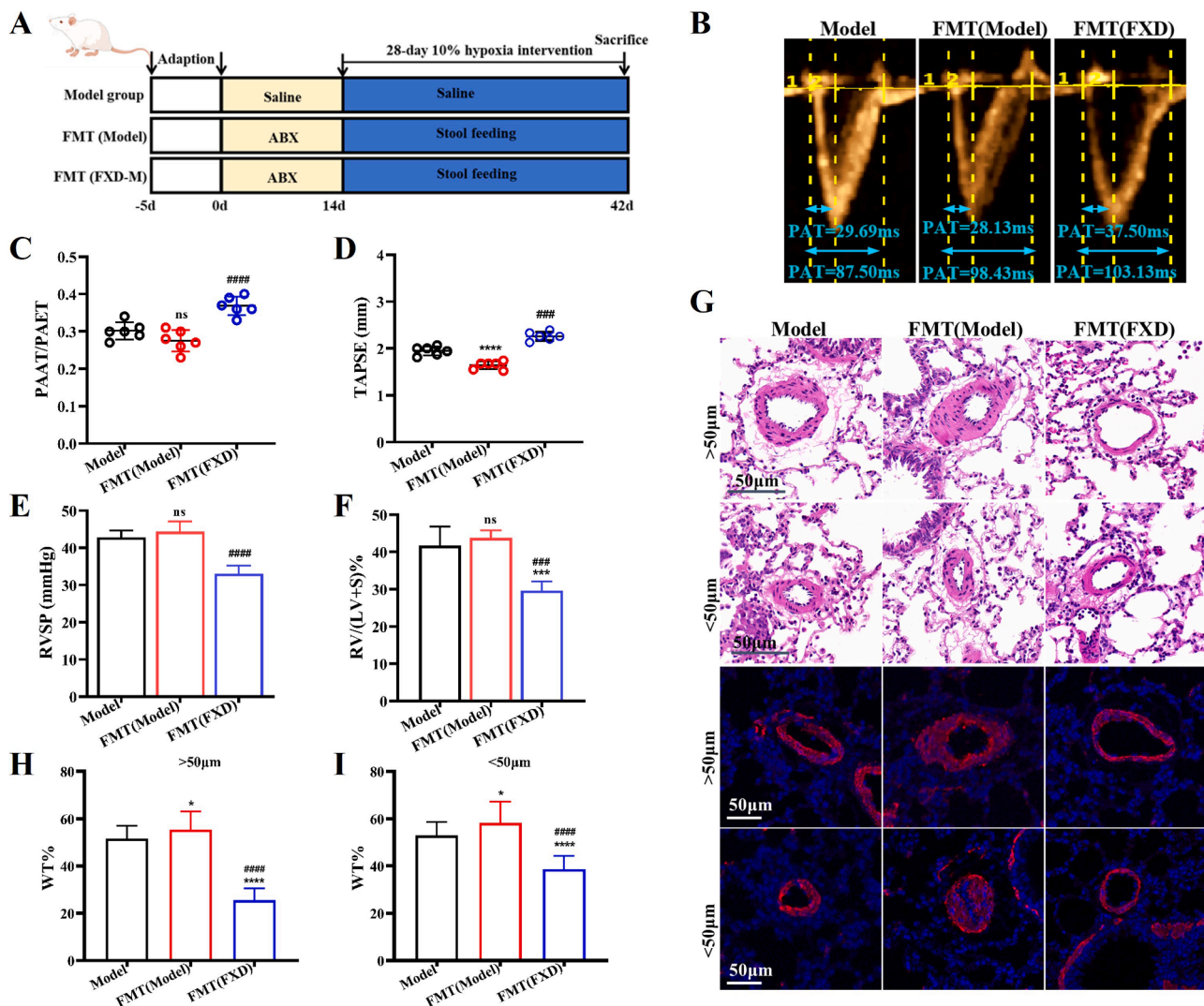


Fig. 8. The effect of FXD mediated by gut microbiota on pulmonary vascular remodeling phenotype in HPH rats A: Experimental design (6 rats per group), ABX administered by gavage for 14 days. B: Echocardiographic measurements of PA function showing PAT/PET ratio; PAT/PET ratio for PAT / PET; C-D: Statistical analysis of PAT/PET and TAPSE; E: Statistical analysis of RVSP; F: Statistical analysis of right ventricular hypertrophy index; G: H&E staining and α -SMA (red) immunofluorescence staining images of PA in HPH rats following FMT transplantation; H-I: Quantitative statistics of PA vascular wall thickness. Compared with the model group, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$; compared with the FMT (model) group, ### $P < 0.001$, #### $P < 0.0001$; ns $p > 0.05$.

The mechanism by which FXD alleviates HPH based on gut microbiota involves TLR4/MyD88/NF- κ B signaling pathway inhibition

FXD improves the intestinal barrier function and the proinflammatory factor expression in plasma, and it regulates intestinal bacteria producing LPS, such as *Escherichia-Shigella* and *Helicobacter pylori*. These factors are closely related to TLR4/MyD88/NF- κ B pathway activation (Hu et al., 2020; Zhao et al., 2021). Thus, we hypothesize that the anti-PH effect of FXD related to gut microbes might be linked to the TLR4 signaling pathway.

The expressions of TLR4, MyD88 and NF- κ B in lung tissue of model group were significantly higher than those of control group ($p < 0.01$), but FXD-M was significantly improved after treatment ($p < 0.01$) (Fig. 9A-D). The above experimental results indicate that FXD therapy inhibits TLR4/MyD88/NF- κ B signaling pathway activation in the lung tissue of HPH rats. As shown in Fig. 9A-D, the expression levels of TLR4, MyD88, and NF- κ B were markedly elevated in the ABX+FXD group compared to the FXD-M group. This suggests that depletion of intestinal microbiota is linked to the activation of the TLR4/MyD88/NF- κ B signaling pathway in lung tissues of rats treated with FXD for HPH.

The lung tissues of the FMT-model and FMT-FXD groups were

extracted for protein experiments to further study the anti-PH mechanism of FXD in HPH rats mediated by gut microbiota. Western blot analysis showed that the expression levels of TLR4, MyD88, and NF- κ B in the lung tissues of FMT-model rats were elevated compared to the model group but reduced compared to FMT-FXD rats (Fig. 9E-H). The above data indicate that the anti-HPH mechanism of FXD based on gut microbiota involves TLR4/MyD88/NF- κ B signaling pathway inhibition.

The microbial metabolites attenuates hypoxia-induced HPASMCs proliferation, migration and resistance to apoptosis

The aberrant proliferation and migration of HPASMCs are key pathophysiological contributors to pulmonary vascular remodeling in PH. Based on our metabolomic analysis revealing significant alterations in microbial metabolites—including EPA, DHA, and GABA—following FXD treatment (Fig. 5), we investigated their effects on HPASMC dysfunction. Primary HPASMCs were cultured and validated, exhibiting >90 % purity (Fig. 10A). Initial CCK-8 proliferation assays determined the optimal treatment concentrations: 100 μ M for EPA and DHA, and 200 μ M for GABA (Fig. 10B-D). Subsequent experiments employed these concentrations. EdU incorporation assays demonstrated that EPA, DHA,

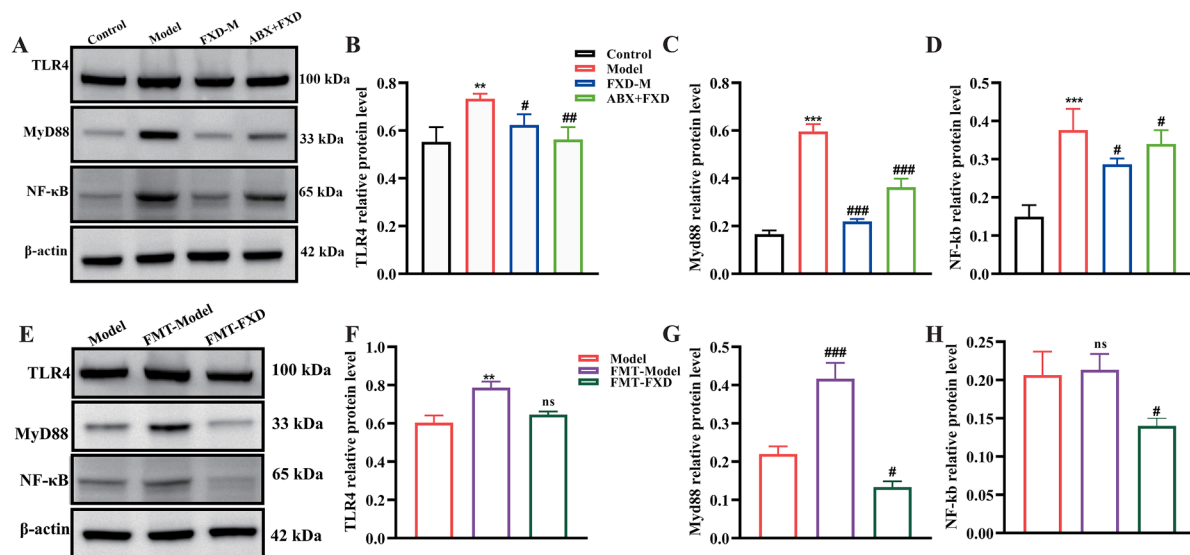


Fig. 9. The anti-HPH mechanism of FXD based on gut microbiota involves the inhibition of the TLR4/MyD88/NF-κB signaling pathway. A-D: Western blot analysis of TLR4, MyD88, and NF-κB protein expression levels in lung tissues from four groups, with quantification of their relative expression intensities. E-H: Western blot analysis of TLR4, MyD88, and NF-κB protein expression levels in lung tissues from three groups, with quantification of their relative expression intensities. Compared to the control group, * $p < 0.05$, ** $p < 0.01$; Compared to the model group, # $p < 0.05$, ## $p < 0.01$; $^{ns}p > 0.05$.

and GABA significantly attenuated hypoxia-induced HPASMC proliferation (Fig. 10E-F). Similarly, transwell migration (Fig. 10G-H) and wound-healing assays (Fig. 10I-J) confirmed that all three metabolites suppressed hypoxia-driven HPASMC migration. Given the role of apoptosis resistance in PH-associated vascular remodeling, we further examined these metabolites' effects using Annexin V-FITC apoptosis assays. Notably, EPA, DHA, and GABA treatment enhanced HPASMC apoptosis under hypoxic conditions compared to controls (Fig. 10 K-L). Collectively, these findings demonstrate that EPA, DHA, and GABA cooperatively mitigate pathological HPASMC proliferation and migration while restoring apoptosis sensitivity, suggesting their role in counteracting pulmonary vascular remodeling in PH.

To further elucidate the mechanistic role of FXD in treating HPH via gut microbiota-derived metabolites, we performed targeted metabolomics analysis revealed significantly higher plasma concentrations of EPA, DHA, and GABA in the medium-dose FXD group compared to the hypoxia group (Fig. S2). When compared with effective in vitro concentrations, the actual plasma EPA level in FXD-treated rats nearly reached the effective concentration (26.16 ± 2.54 vs. $30.245 \mu\text{g/ml}$), while the DHA concentration achieved 38 % of the effective level (12.7 ± 1.49 vs. $32.849 \mu\text{g/ml}$). In contrast, the GABA concentration remained substantially below the effective threshold (0.069 ± 0.007 ng/ml vs. $29.032 \mu\text{g/ml}$) (See Table S4 for unit conversion). These findings demonstrate that FXD improves HPH-associated pulmonary vascular remodeling by regulating beneficial gut microbiota-derived metabolites, particularly EPA and DHA. This provides pharmacokinetic validation for the proposed gut-lung axis mechanism.

The microbial metabolite EPA attenuates hypoxia-induced hyperproliferation of HPASMCs by suppressing the TLR4/MyD88/NF-κB signaling pathway

TLR4 is recognized as a crucial receptor for recognizing LPS, which subsequently recruits its downstream adaptor molecule MyD88 to initiate TLR4-mediated signaling cascades. Our previous in vivo studies demonstrated consistent expression patterns among TLR4, MyD88, and NF-κB, suggesting that TLR4 activation leads to subsequent MyD88/NF-κB pathway activation. To further investigate this mechanism, we treated HPASMCs with the TLR4 agonist LPS to examine MyD88/NF-κB pathway activation. First, we use CCK8 analysis found that

concentrations of LPS (0–10 $\mu\text{g/ml}$) did not affect cell viability (Fig. 11A). Next, we assessed TLR4 transcriptional levels in HPASMCs treated with varying LPS concentrations, observing a dose-dependent increase in TLR4 expression (Fig. 11B). Based on these findings, we selected 5 $\mu\text{g/ml}$ LPS for subsequent experiments. As EPA emerged as the most significantly altered metabolite following FXD treatment, we further investigated its effects on LPS-induced MyD88/NF-κB pathway activation. Fig. 11C-D demonstrates that EPA significantly attenuated LPS-induced abnormal proliferation in HPASMCs. Concurrently, we evaluated EPA's effects on TLR4, MyD88, and NF-κB expression at both transcriptional and protein levels, finding that EPA markedly reduced their expression (Fig. 11 E-G,I-L). Furthermore, we examined whether EPA influences NF-κB nuclear translocation and found that it effectively suppressed LPS-induced nuclear translocation of NF-κB (Fig. 11H).

Discussion

This study first revealed the beneficial therapeutic effect of FXD in HPH, with the FXD-M group demonstrating the best treatment outcomes following comprehensive efficacy assessments. Importantly, the FXD dosage that exerts therapeutic effects is both safe and well-tolerated. This study reveals the association between gut microbiota and FXD efficacy in the HPH model. Furthermore, the potential mechanism of FXD's anti-PH effect is associated with microbiota-mediated inflammatory responses.

Considering that intestinal microbial imbalance contributes to the pathogenesis and progression of HPH (Chen et al., 2022), we investigated the interplay between FXD and gut microbiota. 16S rRNA sequencing analysis revealed that FXD treatment significantly increased the abundance of beneficial bacteria, including *Lactobacillus*, *Alloprevotella*, *Lachnospiraceae_NK4A136*, and *Romboutsia*, while reducing pathogenic bacteria such as *Shigella* and *Helicobacter*. *Lactobacillus*, a well-known probiotic, enhances innate and adaptive immunity, suppresses pro-inflammatory mediators, exhibits broad-spectrum antimicrobial activity, and participates in butyrate metabolism and unsaturated fatty acid biosynthesis (Mo et al., 2022). Meanwhile, *Alloprevotella*, *Lachnospiraceae_NK4A136*, and *Romboutsia* are short-chain fatty acid (SCFA)-producing bacteria that regulate lipid metabolism and mitigate intestinal inflammation (Ma et al., 2020; Montgomery et al., 2024; Ye et al., 2021). Notably, reduced levels of SCFA-producing

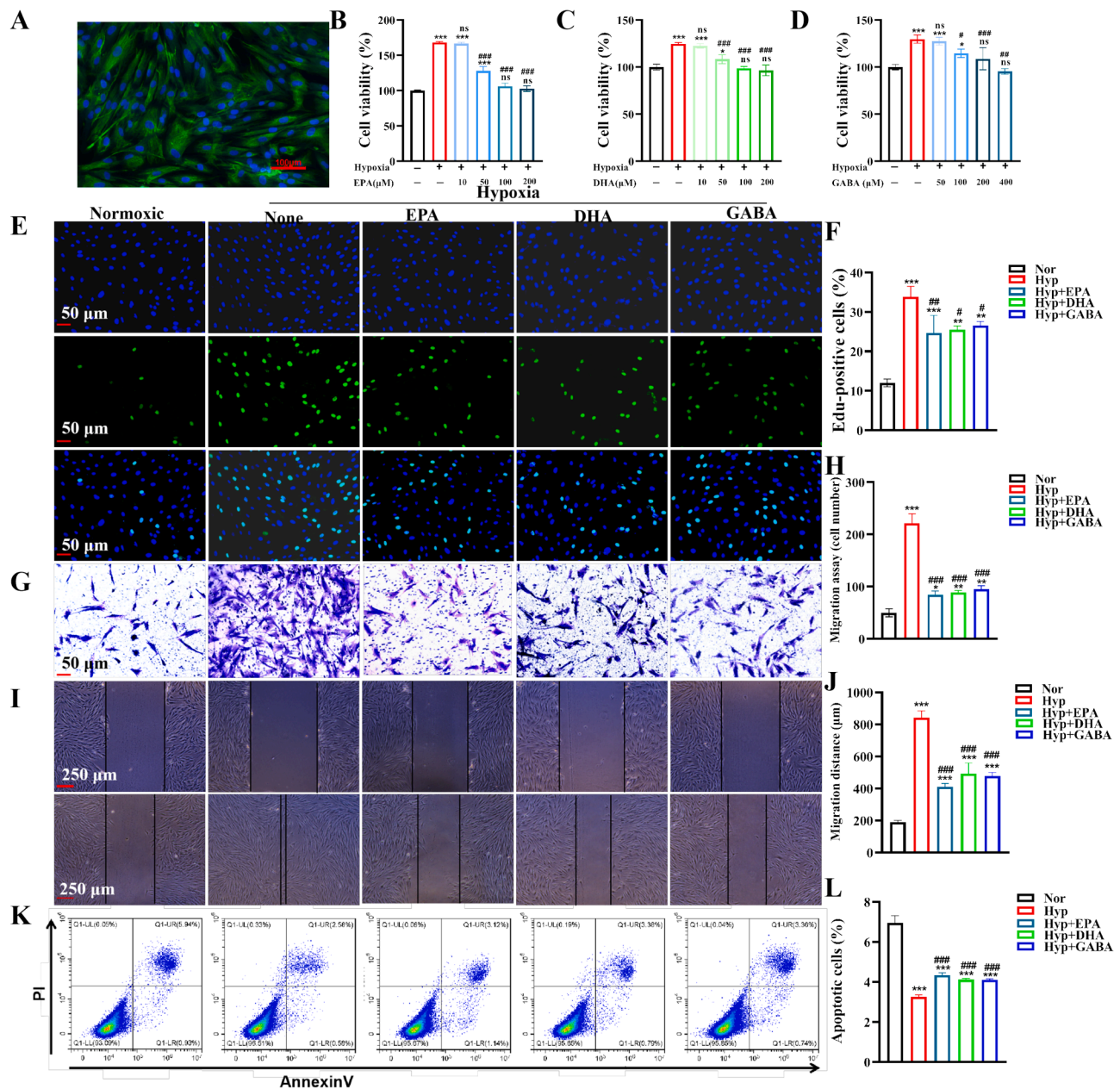


Fig. 10. Effects of microbial metabolites on hypoxia-induced hPASCs proliferation, migration, and apoptosis. A: Purity assessment of hPASCs using α -SMA immunofluorescence staining (green). B-C: The proliferation of hPASCs treated with vehicle or EPA (at dose of 10,50,100,200 μ M), DHA (at dose of 10,50,100,200 μ M) or BABA (at dose of 50,100,200,400 μ M) according to CCK-8 assay ($n = 6$). E-F: The proliferation of hPASCs treated with EPA (100 μ M), DHA (100 μ M) or BABA (200 μ M) according to EdU incorporation assay ($n = 3$). G-H: The migration ability of hPASCs in each group by transwell assay ($n = 3$). I-J: The wound-healing assay of each group ($n = 3$). K-L: Apoptosis of cells was detected by annexin V staining, and the percentages of apoptotic cells were counted ($n = 3$). The data are presented as the means $\pm$ SEMs. Compared to normoxic group, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Compared to hypoxia group, # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$, ns $p > 0.05$.

bacteria have been observed in both PH patients and HPH rat models (Cao et al., 2024; Moutsoglou et al., 2023). Conversely, *Escherichia coli*, *Shigella*, and *Helicobacter* are LPS-producing bacteria (Hu et al., 2020; Zhao et al., 2021). LPS is known to activate TLR4 signaling, triggering systemic inflammation and exacerbating PH (Wang et al., 2019). In summary, FXD-induced modulation of gut microbiota may play a key role in preventing and treating HPH. Since FXD primarily enriches *Lactobacillus*, future studies should explore whether commercially available *Lactobacillus* strains can similarly influence PH development.

Metabolomic analysis demonstrated significant alterations in the metabolic profile following FXD treatment, with particularly notable modulation of unsaturated fatty acids including EPA and DHA. These

bioactive lipids exhibit well-established anti-inflammatory properties and demonstrate vasculoprotective effects in PH. Kurahara et al. documented that EPA supplementation ameliorated PH symptoms in monocrotaline (MCT)-induced rat models (Kurahara et al., 2020). Parallel studies revealed that DHA administration attenuated PH progression in both hypoxia-induced and MCT-induced PH models (Chen et al., 2018; Yan et al., 2013). Clinical investigations further corroborated these findings, showing DHA-mediated improvement in pulmonary vascular tone and reduction of pulmonary arterial pressure in PH patients (Nagaraj et al., 2016). Short chain unsaturated fatty acids (SCFAs), especially butyric acid, is another metabolite that can block the progress of PH (Karoor et al., 2021). Notably, FXD treatment significantly

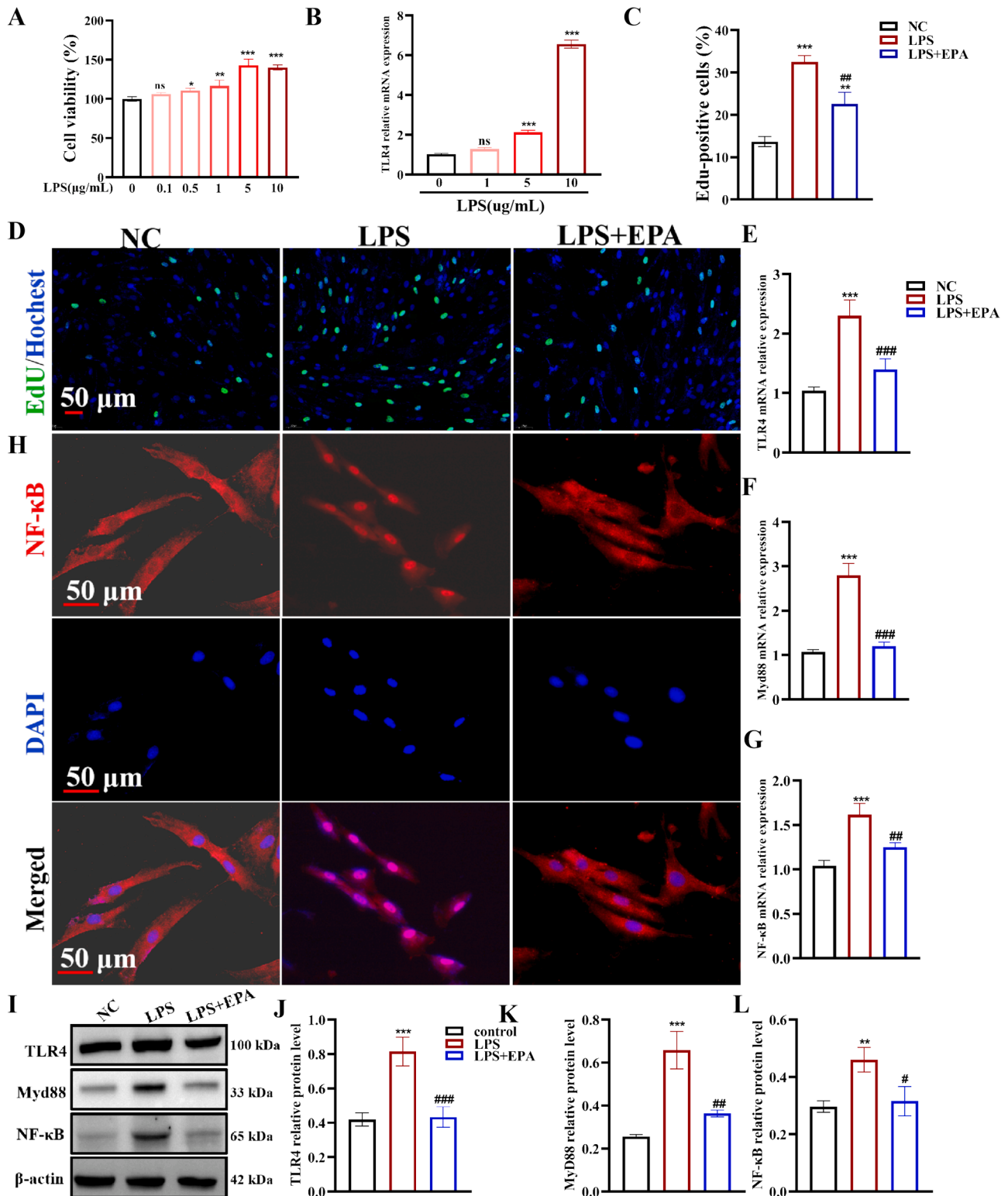


Fig. 11. The microbial metabolite EPA attenuates hypoxia-induced HPASMC proliferation via TLR4/MyD88/NF-κB pathway inhibition. **A:** Viability of cells treated with different concentrations of LPS. **B:** qRT-PCR assay showed TLR4 expression after 0,1,5,10 μg/ml LPS stimulation for 24 h in HPASMCs. **C-D:** The proliferation of HPASMCs treated with LPS (5 μg/ml), or LPS (5 μg/ml)+EPA (100 μM) according to EdU incorporation assay ($n = 3$). **E-G:** qRT-PCR assay showed TLR4,Myd88,NF-κB expression in each group. **H:** Immunofluorescence analysis of the localization of endogenous p65 (red) in HPASMCs after LPS or LPS+EPA stimulation for 24 h. **I-L:** Western blot analysis of TLR4, MyD88 and NF-κB protein expression levels in HPASMCs from three groups, with quantification of their relative expression intensities. Compared to NC group, * $p < 0.05$, ** $p < 0.01$,*** $p < 0.001$, $^{ns}p > 0.05$. Compared to LPS group, # $p < 0.05$, ## $p < 0.01$,### $p < 0.001$.

elevated GABA levels. Subsequent in vitro experiments showed that EPA, DHA and GABA could inhibit the pathological behavior of HPASMCs, including excessive proliferation, enhanced migration ability and anti-apoptosis. Targeted metabolomics found that the EPA concentration in FXD-M group reached the level of biological activity observed in vitro, indicating that it played a key role in FXD-M-mediated metabolite regulation. Of particular interest was the observed significant reduction in serotonin levels post-FXD treatment. This finding holds considerable clinical relevance, as substantial evidence has established the pathogenic roles of both serotonin and trimethylamine N-oxide (TMAO) in promoting PH development (Dempsey et al., 2008; Huang et al., 2022; Yang et al., 2022). Due to the need for targeted metabolomics to measure TMAO, the effect of FXD on TMAO levels cannot be evaluated in our study. Collectively, these findings suggest that FXD exerts its therapeutic effects on HPH, at least in part, through modulation of key metabolic pathways. The demonstrated alterations in EPA, DHA, GABA and serotonin levels indicate a multi-target regulatory mechanism underlying FXD's pharmacological activity. Future investigations should prioritize: (1) elucidating the precise molecular mechanisms by which FXD regulates these critical metabolites, and (2) exploring potential synergistic effects among these metabolic changes. Such mechanistic studies will facilitate the development of more targeted and effective therapeutic strategies for HPH management.

Pulmonary inflammation has been recognized as a critical contributor to hypoxia-induced pulmonary vascular remodeling (Stenmark et al., 2006), highlighting the therapeutic significance of anti-inflammatory strategies in HPH management. Our findings reveal that FXD significantly inhibits circulating pro-inflammatory cytokines. Notably, model rats exhibited increased abundance of LPS and LPS-producing bacterial taxa, which were markedly attenuated by FXD intervention. Previous studies have demonstrated upregulation of the LPS receptor TLR4 and its downstream mediator NF- κ B in both HPH rodent models and COPD patients with PH (Wedgwood et al., 2020; Young et al., 2010). Additionally, hypoxic TLR4-deficient mice show neither significant alterations in pulmonary IL-1 β /TNF α expression nor substantial vascular remodeling (Young et al., 2010), suggesting TLR4 activation mediates the pathophysiological link between gut microbiota dysbiosis and HPH development.

The TLR4/MyD88/NF- κ B pathway emerges as a key inflammatory axis regulated by gut microbiota. Fecal microbiota transplantation has been shown to ameliorate rotenone-induced Parkinson's disease in mice through inhibition of the TLR4/MyD88/NF- κ B pathway (Zhao et al., 2021). Similarly, Xiaochaihu decoction modulates gut microbiota-mediated TLR4/MyD88/NF- κ B signaling to attenuate tumor growth in cancer-comorbid depression (Shao et al., 2021). Complementing these findings, pharmacological inhibition of TLR4 with TAK-242 has been demonstrated to effectively reduce PH and inflammation (Wedgwood et al., 2020). Our experimental results indicate that the expression of TLR4, MyD88, and NF- κ B is reduced in both FXD and FMT-FXD treatment groups. The above results indicate that the specific mechanism of FXD in treating PH involves TLR4/MyD88/NF- κ B pathway activation, which is partly dependent on the action of gut microbiota.

Chemical compounds identified from FXD, such as Astragaloside IV, have demonstrated significant potential in ameliorating pulmonary vascular remodeling and inhibiting the PASMCs (Li et al., 2022). Notably, Liu et al. revealed that Astragaloside IV not only modulates the abundance of beneficial bacteria, including *Lactobacillus*, but also helps restore the balance of SCFAs, highlighting its dual role in gut microbiota regulation and metabolic homeostasis (Y Liu et al., 2023). Ginsenoside, another major component of FXD, has been shown to reduce pulmonary artery pressure and suppress PASMC proliferation in HPH rats by mitigating inflammation (Tang et al., 2023). Beyond its anti-inflammatory properties, ginsenoside exerts protective effects against hypoxia-induced fibrosis and PASMC proliferation, further underscoring its therapeutic potential (Ran et al., 2024). Quercetin, a key bioactive

compound in FXD, has been found to inhibit hypoxia-induced PASMC proliferation and migration while promoting apoptosis (He et al., 2015). Additionally, quercetin downregulates inflammatory responses and improves pulmonary vascular remodeling in HPH (Gao et al., 2024). Further studies have demonstrated that quercetin alleviates LPS-induced intestinal inflammation and enhances gut function by increasing the abundance of SCFA-producing bacteria and promoting the growth of beneficial microbial taxa (Feng et al., 2023). Despite these promising findings, further research is necessary to fully elucidate the anti-HPH efficacy and underlying mechanisms of the primary chemical components in FXD, particularly their interactions with gut microbiota and inflammatory pathways.

It is important to note that while this study investigated the correlation between plasma metabolites and gut microbiota in the context of FXD, the intricate interactions between these factors remain incompletely understood and warrant further exploration. Future research should prioritize identifying the specific gut microbiota taxa modulated by FXD and their associated plasma metabolites, as well as elucidating their functional roles in the pathogenesis and progression of HPH. Furthermore, this study did not address the primary chemical components of FXD or the potential synergistic effects of its individual bioactive compounds on HPH. Investigating these aspects could provide deeper insights into the mechanisms underlying FXD's therapeutic efficacy and represents a critical avenue for future research.

In summary, our study demonstrates that FXD effectively alleviates HPH and restores intestinal ecological balance. The findings reveal that FXD modulates gut microbiota composition, suppresses the activation of the TLR4/MyD88/NF- κ B signaling pathway, regulates inflammatory cytokine levels, and enhances intestinal barrier function, as illustrated in Fig. 12. These results highlight FXD as a promising therapeutic candidate for the clinical treatment of HPH, offering a multifaceted approach to addressing both pulmonary and intestinal dysregulation associated with the disease.

Conclusion

In conclusion, our study provides robust evidence supporting the role of FXD as a potent modulator of gut microbiota. The decoction effectively alleviates symptoms of HPH by restoring gut microbiome balance and modulating associated pathological pathways. These findings not only underscore the therapeutic potential of FXD but also propose a novel, microbiota-targeted strategy for the management of HPH, offering new insights into its treatment paradigm.

Declaration of generative AI in scientific writing

In this study, we did not use any generative artificial intelligence tools or technologies. All research design, data analysis, and manuscript writing were conducted independently by the authors to ensure the originality and academic integrity of the research. If you have any questions, please contact the authors.

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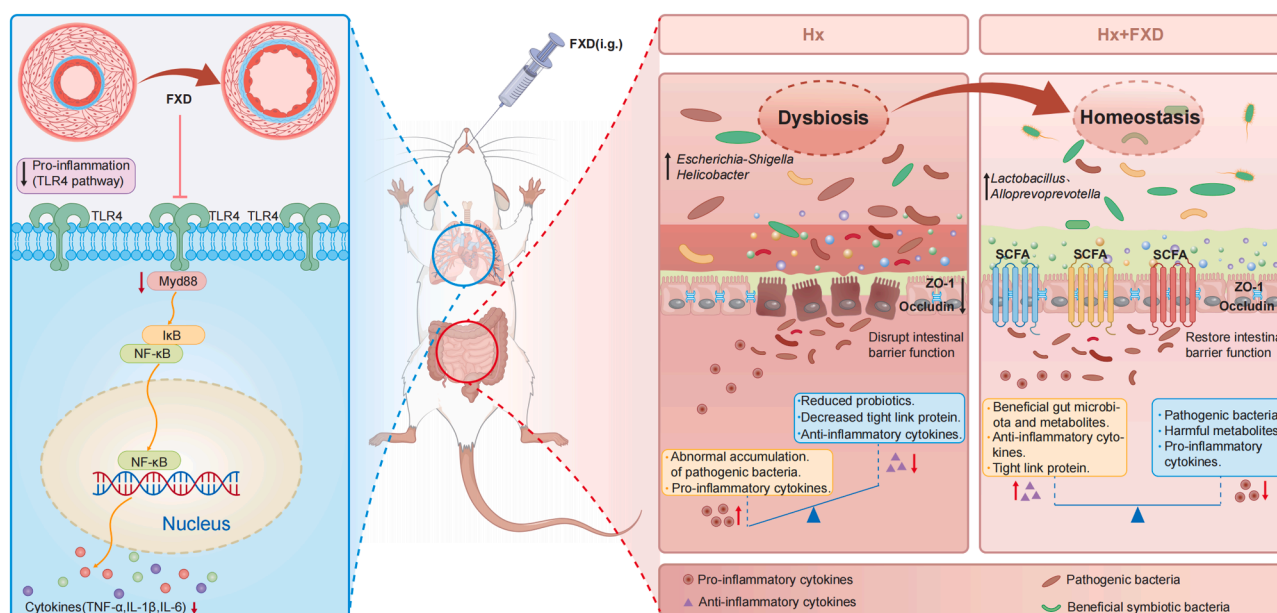


Fig. 12. Schematic diagram of the mechanism. FXD demonstrates the progression of HPH by modulating the TLR4/MyD88/NF-κB signaling pathway mediated by gut microbiota.

CCRediT authorship contribution statement

Fei-ying Wang: Writing – original draft, Methodology, Investigation, Formal analysis. **Jian Yi:** Validation, Data curation. **Ling-ling Zhou:** Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Jun-lan Tan:** Writing – review & editing, Supervision, Resources, Methodology. **Xian-ya Cao:** Validation, Methodology. **Chao Zhang:** Validation, Methodology, Investigation. **Jia-jing Wan:** Validation, Methodology, Data curation. **Lan Song:** Writing – review & editing, Resources. **Ai-guo Dai:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.phymed.2025.157118](https://doi.org/10.1016/j.phymed.2025.157118).

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